



DEFORESTATION IN NEPAL

What the Data Shows and What Can Be Done

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Submitted to

Coding For Social Good Nepal

Summer Camp 2026

September 2026

Abstract

Nepal's forests have changed significantly over the past century. Forest management has moved from centralized state control focused on resource extraction toward community-based governance oriented around conservation. New problems have come up because of building roads, farming land, fires and people moving around in different ways. This report examines these changes using published articles/literatures and recent satellite imagery.

Reddy et al. (2018) found that Nepal's forest cover declined from about 76,710 km² in 1930 to 39,392 km² in 2014, a drop of half. Recent studies show that the loss of trees is not the same everywhere. Some places have problems with losing trees, especially near roads and places where big projects are happening. Between 2001 and 2016 Nepal lost about 46,000 hectares of trees while 12,200 hectares of trees grew back (Shrestha et al., 2018).

The problems from losing trees go beyond just having trees. Breaking up the forests makes it harder for animals to live there. Changing how people use the land and the forests also makes it harder to find water. Makes it harder for people in the countryside. In the Rangun Khola area for example 73% of the places where water comes out of the ground have less water coming out (Pandit et al., 2024).

Nepal has a set of rules to help manage forests, with local communities playing an important role through community forestry. However, challenges remain in implementing policies, building institutional capacity, ensuring inclusive participation and monitoring forest management (Laudari et al., 2019; Kanel & Kandel, 2004; World Bank, 2025). Meanwhile, satellite-based tools such as Global Forest Watch (now Global Nature Watch) and GLAD Alerts provide faster ways to identify possible forest disturbances (GFW Blog, 2020; NASA Science, 2026).

Taken together, these findings show the need to connect local forest management with newer monitoring tools. Satellite-based systems can help identify changes in forest cover and detect areas of concern, while local communities and people working on the ground can provide the context needed to understand and respond to those changes. The report therefore highlights the importance of protecting water sources, using forest and land-use maps when planning roads and other projects, strengthening local monitoring, and making forest-management processes more inclusive and accountable.

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1. Introduction

Nepal's forests are very important to the country's nature and economy. They provide habitat for wildlife, support rural livelihoods, regulate water and soil, and store carbon. At the same time, forests face pressure from both human activities and environmental changes. Farming land, building roads, fires and people moving around are all making things harder for the forests in different parts of the country.

Forest change in Nepal is not just about whether the total amount of trees going up or down. Numbers for the country can hide differences between regions and places. An area can retain its classification as forest while simultaneously shrinking in size and becoming increasingly fragmented. It can lose the trees that cover the ground and be more likely to have problems again. Because of that, knowing where the trees are disappearing and what is causing it is as important as knowing the numbers for the country.

This report looks at how the forests have changed over time and recently the main reasons for these changes and what happens because of them to animals, water, soil and people. It also examines the policies that help manage the forests, including people taking care of the forests themselves and a program called REDD+. It also looks at how pictures from satellites can help with the work that is already going on. Finally the report suggests ways to use better checking with action that happens in the local areas.

2. Forest Change in Nepal

2.1 Definitions and Historical Change

Different studies use ways to analyse how forests change so it is important to know the difference between forest cover, tree cover, deforestation and forest degradation. Forest cover usually means land that is considered a forest. Tree cover means the trees and plants in an area. Global Forest Watch, for example, uses pictures from satellites to measure tree cover based on the canopy height. Deforestation means when the forest is changed into something like a farm or a city. Forest degradation means when the forest is not as good as before. It is still a forest. These differences matter because losing some trees does not always mean the forest is gone forever. Losing trees from pictures can come from causes like fires cutting down trees or changing the land to something else. Eaturu (2025) says it is important to know these differences when looking at the numbers about how forests change.

2.1.1 Historical Forest Change

Nepal experienced significant forest loss throughout the twentieth century. Reddy et al. (2018) found that the country had around 76,710 km² of forests in 1930. By 2014 this had dropped to about 39,392 km², which is a decline of about half.

The way the forests changed also got worse. Of just losing trees the forests became split into smaller pieces. Reddy et al. (2018) said the number of forest areas went up from 6,925 in 1930 to 42,961 in 2014. At the same time, large contiguous core forest areas declined substantially. Recent studies show something different from the old numbers. The 2021 National Forest Inventory says that about 40.36% of Nepal's land is covered by forests. This means that the overall trend has changed a lot from the losses that happened before.. Even with this the changes are not over. Loss and problems with the trees still happen in some areas.

2.2 Recent Forest-Loss Trends and Spatial Patterns

Pictures from satellites give a look at what has happened more recently. Shrestha et al. (2018) said that Nepal lost about 46,000 hectares of trees between 2001 and 2016 while 12,200 hectares of trees grew back.

The amount of loss each year changed a lot during this time. The most trees lost in one year was 6,180 hectares in 2009 while the least was about 1,040 hectares in 2015.

The way the changes happened also differed in parts of the country. This means that the changes are not the same in Nepal. Some areas had a lot of tree loss. Shrestha et al. (2018) named Kailali, Dang, Sarlahi, Rautahat and Nawalparasi as places with a lot of loss. These patterns are important because they show where more checking and helping with the trees should happen.

Population pressure is also a problem. Eaturu (2025) found that the national link between how people there are and the loss of trees was not strong with a correlation coefficient of only 0.00029. Some areas in Madhesh Province showed a much stronger local link. In places like Bara and Rautahat the local numbers were 0.75 to 0.81. This shows that the big national numbers can hide what is happening in places.

Changes in how the trees look are also a problem. The World Bank (2025) said that the average coverage of the tops of the trees went down from 67% in 2018 to 64% in 2022. This means that the health of the trees should be checked along with how trees there are when looking at Nepal's forests.

2.3 Major Drivers of Change

The loss of trees in Nepal has reasons and these reasons are different in different places.

Building roads is one of the patterns found in recent studies. Shrestha et al. (2018) found that most tree cover loss and gain occurs within 5 kilometers from roads. The further from roads, the less change is recorded. So the big roads and the places where big projects are happening are areas for checking the trees. Roads can hurt the trees directly by clearing the land and indirectly by making areas that were hard to get to easier to reach. In the middle of the hills, roads that are not planned well can also cause problems with the sides of the hills which makes it more likely to have landslides.

This does not mean that every road project causes loss of trees but the link between the roads and the trees shows why these places should be watched closely.

More farming is another reason for the changes. Turning forests or other land into fields can reduce the number of trees, especially where people need land for farming. Using wood for

energy is also a long-time problem. FAO (2004) said that wood was a part of the energy people used and Shrestha et al. (2018) also said that people living in the countryside still depend a lot on the trees for their needs.

Fires are another cause of problems. Parajuli et al. (2015) using pictures from satellites from 2000 to 2013 counted 2,159 fires each year on average. The fires mostly happened between January and June with the happening in April.

The most fires in a year was in 2012 with 3,518 fires while 2009 had the biggest area burned, around 152,408 hectares. These two years should not be seen as the same when it comes to fires. The research shows that 2012 had the fires but 2009 had the most land burned in the time studied. Our study also found that fires were a reason for losing trees in those years.

Sometimes the changes are not just going one way. Rural out- migration from various country sides of Nepal has led to abandonment of marginal agricultural land in some areas. Shrestha et al. (2018) found that some of these areas have trees growing back naturally. This shows that moving away can have effects on the land depending on where it is and what the land was used for before.

3. Impacts of Forest Loss

Forest loss has consequences that extend well beyond changes in tree cover changing how the trees are arranged and how people use the land can affect where animals live, how water there is, how safe the soil is and how people who depend on these things are affected. The problems are connected, not separate.

3.1 Biodiversity and Habitat Fragmentation

One of the significant consequences of historical forest loss has been fragmentation. Reddy et al. (2018) Found that Nepal's forest patches increased from 6,925 in 1930 to 42,961 in 2014. During the period large core forest areas declined by approximately 75.5%.

Fragmentation can be particularly serious for species that require connected habitats. The Asian elephant is one example. Ram et al. (2021) examined changes in elephant habitat in Nepal between 1930 and 2020 and found that approximately 21.5% of elephant habitat in the Chure-Terai-Madhesh landscape had been lost since 1930.

The effects are not limited to wildlife populations themselves. As forests are converted or fragmented animals can increasingly come into contact with land and settlements. Ram et al. (2021) reported that 60% of recorded elephant attacks on humans between 2000 and 2020 occurred in areas where forests had recently been converted to agriculture or settlements. This illustrates how habitat loss directly contributes to a human-wildlife conflict issue.

3.2 Water Scarcity and the Coping Burden

The relationship between forests, land use and water availability is especially important in Nepal's mid-hills, where many communities depend on springs for water.

Pandit et al. (2024) mapped 1,122 springs in the Rangun Khola watershed. Found that 73% showed declining flow trends while approximately 2% had already dried up. The study also

found changes in forest and agricultural land use. Identified a statistical association between forest decline and changes in spring flow.

Water scarcity creates a significant burden on rural households. Gurung et al. (2019) documented water-related difficulties in watersheds of Nepal's mid-hills and found that local communities perceived significant reductions in available water resources. These changes can increase the amount of time households need to spend collecting water and can place a particularly large burden on women and girls who are often responsible for domestic water collection.

The water issue is therefore closely connected to forest management. Protecting forest areas that contribute to groundwater recharge and spring systems can provide benefits beyond forest conservation itself.

3.3 Soil Degradation and Climate Impacts

Forest vegetation also plays a role in protecting soil particularly on Nepal's steep terrain. When vegetation is removed or disturbed rainfall can easily reach exposed soil and increase erosion. Road construction and agricultural expansion can further disturb slopes.

FAO (2004) reported soil-loss estimates of approximately 20–25 tonnes per hectare in Nepal illustrating the scale of the country's erosion problem. Research from watersheds has also found that agricultural expansion at the expense of forest and other land covers can increase soil loss.

Forests are also important for climate regulation because they store carbon. Adhikari et al. (2022) estimated that Nepal's forests hold 1,054.97 million tonnes of carbon. Forest clearing and burning can release some of this stored carbon into the atmosphere while maintaining and restoring forest ecosystems can contribute to climate mitigation.

4. Forest Policy and Governance in Nepal

Nepal's experience with forest governance is important because the country has moved through different approaches to managing its forests. History shows that simply placing forests under state control does not automatically produce better conservation outcomes. In cases local participation and clear management rights have been equally important.

4.1 Historical Evolution of Forest Governance

Nepal's forest policy has changed considerably since the middle of the century. Laudari et al. (2019) describe this development through four periods.

The first was a period of protection from approximately 1950 to 1975. The Private Forest Nationalization Act of 1957 brought forests under state control. Although the intention was to protect forest resources the policy also created tension between the state and local communities that had historically depended on forests. In some areas uncertainty over access encouraged people to extract forest resources before state control became effective.

During the period from approximately 1975 to 1986 the emphasis shifted toward resource creation. Concern over environmental degradation encouraged the government to focus on plantations and forest development through state and Panchayat institutions.

An important shift occurred during the participatory forestry period beginning around 1986. The Master Plan for the Forestry Sector (1989) and the Forest Act of 1993 strengthened the basis for community involvement in forest management. By treating local people only as users of forest resources, the system increasingly recognized them as participants in managing those resources.

Since 2008 forest governance has increasingly been connected with international climate and carbon discussions. REDD+ has introduced ways of thinking about forests in terms of carbon storage, emissions reduction, monitoring and international results-based finance (Laudari et al. 2019).

4.2 Constitutional and Legal Framework

Nepal's transition to federalism further changed the way natural resources are governed. The Constitution of Nepal (2015) established provincial and local levels of government and distributed responsibilities among them.

Forest and environmental management consequently involve institutions at more than one level. The Forest Act 2019 (2076) is one of the legal foundations for current forest management. It provides provisions for categories of forests, including community forests, leasehold forests and other management arrangements.

The Local Government Operation Act 2017 also gives local governments responsibilities related to management including local water sources and watersheds. These legal arrangements create opportunities for forest and water management to be connected at the level.

At the time, overlapping responsibilities between levels of government can create difficulties when institutional roles and capacity are not clearly aligned.

4.3 Roles of the Three Tiers of Government

Under the system forest governance is no longer the responsibility of a single institution.

At this level the Ministry of Forests and Environment is responsible for national-level policy and coordination. The Forest Research and Training Centre plays a role in forest assessment, research and Measurement, Reporting and Verification (MRV).

Provincial governments have responsibilities for coordinating and implementing forest-related activities within their jurisdictions while local governments are increasingly involved in protection, watershed management and local development planning.

This division of responsibility can potentially make forest management more responsive to conditions. However it also creates implementation challenges. Pandit et al. (2024) for example identified weaknesses in institutional capacity, including vacant environmental positions in some municipalities.

Where technical staff are unavailable local governments may struggle to monitor impacts or properly implement environmental requirements for infrastructure projects. This gap between responsibility and actual capacity is important.

Giving an institution responsibility on paper does not necessarily mean that it has the staff, technical knowledge, funding or monitoring systems needed to carry out that responsibility effectively.

4.4 Community Forestry and Social Equity

Community forestry is one of Nepal's significant contributions to participatory forest management. Under this system local Community Forest User Groups (CFUGs) are given responsibilities and management rights over community forests.

According to FECOFUN (2023) and World Bank (2025), more than 22,000 CFUGs manage over 2.4 million hectares of forest and involve more than three million households. The scale of the system demonstrates how deeply community forestry has become part of Nepal's forest governance structure.

Research has also found relationships between community forest management and forest conditions. Shrestha et al. (2018) found lower levels of tree-cover loss in districts with greater community-forest coverage.

However community management does not automatically guarantee participation or benefits. Kandel and Kandel (2004) identified capture as a major challenge, in which more powerful members of communities can dominate decision-making structures. Poorer and marginalized households may consequently have less influence over how forest resources and community funds are managed.

This creates what can be described as a second-generation challenge for community forestry. The question is no longer only whether local communities are involved. Also who within those communities has a meaningful voice and receives the benefits of forest management.

As forest governance becomes connected with carbon finance and other economic benefits, ensuring inclusive decision-making becomes even more important.

4.5 REDD+ and Carbon Governance

REDD+ has added another dimension to Nepal's forest governance system by linking forest conservation with climate finance. The basic idea is to provide results-based payments for reductions in emissions from deforestation and forest degradation.

Nepal has developed systems for measuring forest carbon and establishing reference levels against which changes can be assessed. Adhikari et al. (2022) describes Nepal's progress in developing REDD+ institutions and a Forest Reference Level based on forest and emissions data.

In November 2025 Nepal received its results-based REDD+ payment of US\$9.4 million from the Forest Carbon Partnership Facility for reported reductions of approximately 1.88 million tonnes of carbon emissions between 2018 and 2024 (National REDD+ Centre, 2025).

This development shows the growing connection between Nepal's forest-management system and international climate finance. However carbon finance also raises questions about transparency, benefit sharing, monitoring and institutional capacity.

Results-based payments are meaningful when the underlying measurements are reliable and when the benefits reach the communities and institutions responsible for protecting forests.

For Nepal REDD+ should therefore be viewed not as a source of funding but as another reason to strengthen forest monitoring, transparent governance and equitable participation.

5. Role of Satellite-Based Monitoring

Nepal's terrain makes regular ground-based forest monitoring challenging. Large and remote forest areas can be difficult to patrol while fires, illegal clearing and road construction can develop faster than conventional monitoring systems can respond.

Satellite technology provides a way to identify disturbances over large areas. Global Forest Watch uses satellite observations to provide information on tree-cover change while GLAD Alerts can provide frequent signals of possible canopy disturbance (GFW Blog, 2020).

Satellite monitoring is particularly useful when it is combined with knowledge. An alert can identify a location that requires attention while local officials, forest groups or rangers can investigate what actually happened on the ground.

Radar-based systems offer an additional advantage. Synthetic Aperture Radar (SAR) that can gather information when optical satellite observations are blocked by cloud cover. This is especially important in areas where cloud conditions can block regular pictures.

NASA Science (2026) describes methods combining radar and observations to accelerate the detection of forest loss.

Satellite alerts should not be seen as proof of illegal action. A change that is found can be from fire cutting trees building things, natural events or other reasons. Checking the ground is still needed before making decisions about enforcing rules.

The best job of satellite monitoring is not to replace people working in forests but to help them decide where and when they need to focus their efforts.

6. Policy Recommendations

The information in this report shows that Nepal does not necessarily need a forest management system. Instead current groups can be made stronger by connecting data, local action and more open decision-making.

6.1 Strengthen Local Forest Monitoring Stronger

Satellite warnings should be used in the work of forest offices, rangers and community forest groups. Tools like Forest Watcher can help local teams find areas that need checking.

A good system could involve someone who reviews warnings, checks the areas on the ground and writes down what they find. The goal is to make monitoring smarter, not more frequent. Satellite data can find places while field teams find out what really happened.

6.2 Formalize Springshed Management

Forest policy should pay attention to the link between forests and water. The finding that 73% of mapped springs in the Rangun Khola area had declining flow is a strong reason to link forest protection with managing the area around springs (Pandit et al., 2024).

Actions like protecting areas that refill water, fixing damaged forests upstream and building water storage and refill systems can be added to local plans. These steps should be based on each area's features not used the same everywhere.

6.3 Use Data About Places in Infrastructure Plans

The fact that tree cover loss is concentrated near roads shows that planning for roads and other projects should use more place-based information.

Before approving roads and other projects planners should find nearby places where trees are being lost, areas with no trees, wildlife homes and places that are easy to damage.

Geospatial analysis can help planners compare paths and find places where building could cause big environmental issues. Environmental checks should then use data instead of just general ideas about possible effects.

6.4 Make Community Forestry More Transparent

The success of community forestry should not only be judged by the state of the forest. It should also be seen by how many people take part and how benefits are shared.

CFUGs should make sure that people who have been left out can take part in decisions. Keeping records of community money and payments for carbon can help stop more powerful groups from taking over.

As other money systems become more important communities need to know how money is given out, who gets it and how choices are made.

6.5 Strengthen Local Skills

Federal and local groups can make rules. Many choices about the environment are made locally. So local governments need trained workers, training and money to do the work of checking and planning the environment.

Fixing jobs that are not filled and giving access to geospatial and forest management help can make the gap between rules and what actually happens smaller.

7. Conclusion

Nepal's forest history shows both the extent of forest loss and the possibility of recovery when forests are properly protected and managed. Nearly half of Nepal's forest cover was lost between 1930 and 2014, but the introduction of community forestry under the Forest Act of 1993 marked a turning point in how forests are governed. However, recent satellite-based studies show that forest loss and degradation are still occurring, with problems increasingly connected to roads, agriculture, fires, and other human activities.

The impact goes beyond the amount of forest cover. Forest fragmentation has broken previously connected habitats into isolated patches, while declining forest cover has contributed to drying springs in Nepal's mid-hills, where communities depend on these sources for daily water. Soil erosion on Nepal's steep slopes and the release of stored forest carbon compound these challenges at both the local and global level. This makes forest protection important not only for biodiversity but also for clean water, climate goals, and the people who depend on these resources.

Nepal already has many of the institutions and community groups needed to address these challenges. The greater challenge is improving coordination among them. Satellite monitoring can help identify forest problems early, while local communities can verify these changes and take action. Community forestry can play an important role in protecting forests, but its long-term success requires addressing elite capture and ensuring that marginalized local people have a genuine voice in decisions and a fair share of benefits. Policy without capacity is paper. The gap between what Nepal's forest law requires and what local governments can actually deliver remains one of the most important problems to solve.

A stronger approach would combine reliable data, faster local responses, improved management of watersheds, and more transparent decision-making. Rather than treating technology, policies, and communities as separate solutions, Nepal can bring them together to create a

forest-protection system that is more practical, responsive, and connected to what is happening on the ground. Nepal's forests are not waiting for a perfect policy. They are responding right now to roads being cut into hillsides, to springs running lower each dry season, and to fires burning across the mid-hills every April. The data is clear about where the pressure is greatest and what is at stake. What remains is the will to act on it with the speed, fairness, and local grounding that the situation demands.

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